TRIBHUVAN UNIVERSITY

INSTITUTE OF ENGINEERING

THAPATHALI CAMPUS

A PROJECT PROPOSAL

ON

EXPLAINABLE WIND-AWARE SPATIO-TEMPORAL GRAPH NEURAL NETWORK FOR AIR QUALITY FORECASTING AND MITIGATION ANALYSIS

Submitted By:

Aman Singh	(THA079BCT002)
Avinash Joshi	(THA079BCT007)
Kaushik Rayamajhi	(THA079BCT018)
Nirika Lamichhane	(THA079BCT024)

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Department of Electronics and Computer Engineering

Thapathali Campus

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Aman Singh	(THA079BCT002)
Avinash Joshi	(THA079BCT007)
Kaushik Rayamajhi	(THA079BCT018)
Nirika Lamichhane	(THA079BCT024)

ABSTRACT

Ambient air pollution is a major environmental and public-health concern, especially in urban regions where pollutant concentration changes across both space and time. Traditional forecasting approaches based only on individual monitoring stations often fail to capture spatial pollutant movement caused by wind, distance, and interaction between monitoring locations. This proposal presents an explainable wind-aware Spatio-Temporal Graph Neural Network (ST-GNN) framework for air quality forecasting and mitigation analysis. A city or selected region will be represented as a dynamic graph where air quality monitoring stations are nodes and edges represent spatial relationships influenced by geographic distance, wind direction, wind speed, and historical pollutant correlation. The model will combine graph attention or graph convolution based spatial learning with temporal models such as Gated Recurrent Unit (GRU) or Long Short-Term Memory (LSTM) to forecast pollutants such as $PM_{2.5}$, PM_{10}, NO_2, and O_3 for future horizons. The system will include an explainability module to analyze which neighboring stations, weather factors, and temporal patterns influenced each prediction. A what-if scenario module will also support mitigation analysis by simulating possible local intervention effects. The final output will be an interactive dashboard for station-level forecasting, risk visualization, explanation, and decision support.

Keywords*—Air quality forecasting, spatio-temporal graph neural network, graph attention network, $PM_{2.5}$, PM_{10}, wind-aware graph, explainable artificial intelligence, environmental intelligence.*

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1 INTRODUCTION

1.1 Background

Air pollution is one of the most serious environmental issues affecting public health, urban planning, and quality of life. Pollutants such as $PM_{2.5}$, PM_{10}, NO_2, O_3, CO, and SO_2 can vary significantly over time and across locations. These variations are influenced by traffic, industrial activity, weather, wind direction, wind speed, temperature, humidity, and regional pollutant transport.

Air quality forecasting is therefore a spatio-temporal problem. The pollutant concentration at one monitoring station depends not only on its previous readings but also on neighboring stations, meteorological conditions, and the transport of pollutants from surrounding areas. Machine learning and deep learning models such as LSTM and GRU can learn temporal patterns, but they may not fully capture spatial relationships between monitoring stations when each station is treated independently.

Graph Neural Networks (GNNs) are suitable for learning from relational data. In the proposed project, the air quality monitoring network is represented as a graph. Each monitoring station becomes a node, and the relationship between two stations becomes an edge. By combining graph-based spatial learning with recurrent temporal learning, the system can model both station-to-station influence and pollutant variation over time.

1.2 Motivation

Traditional station-wise forecasting can be insufficient because pollution does not remain fixed around one station. Pollutants can travel from one area to another due to wind and atmospheric movement. A wind-aware graph-based model can provide better insight into how pollution may spread across different monitoring stations and can adapt edge influence when wind conditions change.

Another motivation is explainability. Forecasting models should not only provide predicted pollutant values but also explain why a certain area is expected to have high pollution. By analyzing attention weights, feature importance, wind direction, and neighboring station influence, the proposed system aims to provide human-understandable explanations for air quality predictions.

1.3 Problem Definition

Air quality forecasting is a challenging spatio-temporal problem because pollutant concentration at a monitoring station depends not only on its past readings, but also on surrounding stations, meteorological conditions, and wind-driven pollutant movement. Many existing forecasting approaches either treat monitoring stations independently or use fixed spatial relationships, which may not accurately represent changing pollutant dispersion caused by variations in wind direction and wind speed.

Furthermore, several forecasting models provide numerical predictions without clearly explaining which neighboring stations, weather factors, or previous time steps influenced the forecast. This lack of explainability limits the usefulness of such models for environmental monitoring and decision-support. Existing systems also rarely provide scenario-based analysis to estimate how changes in pollution levels at selected locations may affect surrounding areas.

Therefore, the problem addressed in this project is to design and develop an explainable wind-aware spatio-temporal forecasting framework that can model dynamic relationships between air quality monitoring stations, predict future pollutant concentrations, identify important influencing factors, and support what-if scenario analysis for mitigation planning.

1.4 Objectives

The objectives of the project are:

1. To develop an explainable wind-aware ST-GNN model for air quality forecasting using pollutant and meteorological data.
2. To design an interactive dashboard that visualizes predicted air quality, station-level risk, explanation of model predictions, and what-if mitigation scenarios.

1.5 Scope and Applications

The scope of this project includes air quality forecasting using data from public monitoring stations and meteorological sources. The system will focus on pollutants such as $PM_{2.5}$, PM_{10}, NO_2, and O_3, depending on dataset availability. The forecasting horizon may include short-term predictions such as 1 hour, 6 hours, 12 hours, and 24 hours.

The system will be developed as a research prototype and decision-support tool. It will not replace official government air quality warning systems. The applications of the project include air quality forecasting, environmental monitoring, pollution risk visualization, urban environmental planning support, station-level pollution influence analysis, public awareness dashboards, and what-if mitigation scenario analysis.

2 LITERATURE REVIEW

2.1 Traditional Air Quality Forecasting

Traditional air quality forecasting methods include statistical time-series models, regression models, and interpolation techniques. These methods are useful for simple forecasting problems, but they may not effectively capture complex nonlinear and spatio-temporal pollutant patterns. Since air pollution is influenced by weather, wind, traffic, and regional interactions, station-wise forecasting alone may not be sufficient for accurate regional prediction.

Baseline models remain important in this project because they provide a clear reference point for evaluating advanced graph learning models. Persistence models, linear regression, random forest, Support Vector Machine (SVM), LSTM, and GRU can be used to compare whether the proposed wind-aware ST-GNN provides meaningful improvement.

2.2 Deep Learning for Time-Series Forecasting

Deep learning models such as LSTM and GRU are widely used for time-series forecasting because they can learn temporal dependencies from historical data. In air quality forecasting, these models can learn how pollutant concentration changes over time. However, normal LSTM and GRU models usually focus on temporal patterns and may not fully model spatial relationships between different monitoring stations.

Spatio-temporal forecasting research combines temporal sequence models with spatial models to address this limitation. For example, recent $PM_{2.5}$ forecasting work uses recurrent units with graph neural components to capture both time evolution and spatial diffusion across monitoring locations [1].

2.3 Graph Neural Networks for Air Quality Forecasting

GNNs can model relationships between monitoring stations by representing the monitoring network as a graph. In this graph, stations are nodes and spatial relationships are edges. Spatio-temporal GNN models combine graph learning with temporal models to capture both spatial and temporal dependencies. This makes them suitable for air quality forecasting because pollutant levels at one station can be affected by neighboring stations.

PM2.5-GNN is an example of a domain-knowledge enhanced graph neural model for $PM_{2.5}$ forecasting [2]. E-STGCN further shows that spatio-temporal graph convolutional approaches can be adapted for air pollution forecasting and can address extreme pollutant behavior in polluted cities [3]. These studies motivate the proposed project's use of graph-based spatial learning with temporal forecasting.

2.4 Wind-Aware Dynamic Graph Modeling

Many graph-based forecasting models use fixed edges based on distance or historical correlation. However, pollutant dispersion can change depending on wind speed and wind direction. A wind-aware graph can update edge weights dynamically based on whether wind flows from one station toward another. This can make the graph structure more realistic and physically meaningful.

In the proposed system, edge weights will combine distance, historical pollutant correlation, wind speed, and wind direction alignment. This approach is expected to better represent directional pollutant transport than a purely distance-based graph.

2.5 Explainable AI in Environmental Forecasting

Deep learning models are often considered black boxes. For environmental decision-support systems, explainability is important because users need to understand why a model predicts high pollution in a certain area. In this project, attention weights, feature importance, and neighboring station influence will be analyzed to provide interpretable explanations.

GNNExplainer and related explanation methods show how graph model predictions can be interpreted by identifying important graph structures and input features [4]. For the proposed system, explanation outputs will be compared with wind direction, distance, weather features, and pollutant history to check whether the learned relationships are physically reasonable.

2.6 Research Gap

Existing air quality forecasting methods often focus either on temporal prediction or fixed spatial relationships. A practical gap remains in combining wind-aware dynamic graph construction, explainable prediction, and scenario-based mitigation analysis in one system. This project addresses the gap by integrating public air quality data, meteorological data, spatio-temporal graph learning, explanation, and an interactive dashboard.

Table 2-1 Comparison of Related Approaches and Proposed System

Approach	**Primary Focus**	**Limitation / Difference from Proposed System**
Station-wise regression	Forecasts pollutant concentration for each monitoring station independently.	Does not capture spatial pollutant movement or influence from neighboring stations.
LSTM/GRU-based forecasting	Learns temporal patterns from historical pollutant and meteorological data.	Captures time dependency but may ignore station-to-station spatial influence.
Fixed-graph ST-GNN	Models spatial and temporal dependencies using a predefined graph structure.	Graph edges are usually static and may not adapt to changing wind speed and wind direction.
Wind-aware ST-GNN	Uses distance, wind direction, wind speed, and pollutant correlation to model dynamic station influence.	Provides more physically meaningful edge weighting for pollutant transport modeling.
Proposed system	Combines wind-aware forecasting, explainability, risk visualization, and what-if scenario analysis.	Extends forecasting results into an interpretable decision-support dashboard for air quality analysis.

3 METHODOLOGY

3.1 System Block Diagram/Architecture

The proposed system is organized into data collection, preprocessing, dynamic graph construction, spatio-temporal forecasting, explainability, what-if analysis, backend service, and dashboard layers.

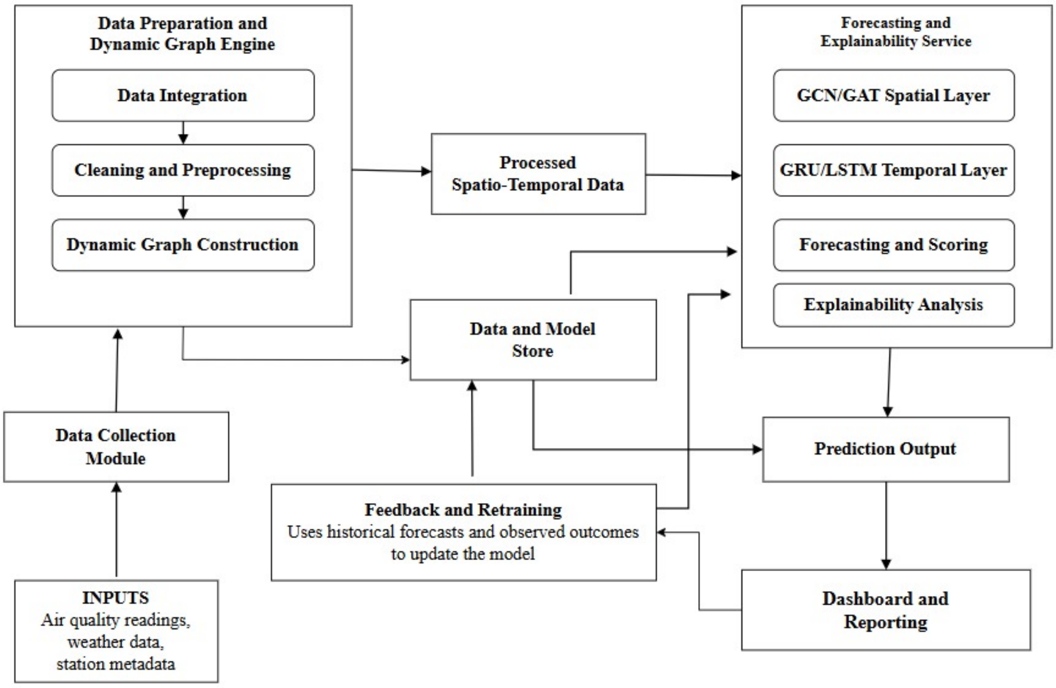

Figure 3-1 System Block Diagram

3.2 Description of Working Principle

The proposed system works as a wind-aware air quality forecasting and decision-support system. It combines air quality data, meteorological data, dynamic graph construction, spatio-temporal deep learning, explainability, and dashboard-based visualization. The overall working principle of the system is described in the following subsections.

3.2.1 Data Collection

The first step of the system is data collection from selected air quality monitoring stations and meteorological data sources. The air quality data consists of pollutant concentrations such as $PM_{2.5}$, PM_{10}, NO_2, SO_2, CO, and O_3. These pollutants are selected because they are commonly used indicators for assessing air pollution levels and public health risk. Along with pollutant values, station-related information such as station location, latitude, longitude, and timestamp is also collected.

In addition to air quality data, meteorological features are collected because weather conditions strongly influence the movement and concentration of pollutants. The meteorological data includes wind speed, wind direction, temperature, humidity, and atmospheric pressure. Among these features, wind speed and wind direction are especially important because they help determine how pollutants may travel from one monitoring station area to another. The collected data may be obtained from open environmental platforms, weather APIs, and publicly available benchmark datasets.

3.2.2 Data Cleaning and Preprocessing

After data collection, the raw data is passed through the preprocessing stage. Since air quality and meteorological datasets are collected from different sources, they may have different timestamp formats, missing readings, inconsistent sampling intervals, and abnormal values. Therefore, timestamp synchronization is performed to align pollutant and weather records according to a common time interval.

Missing values are handled using suitable methods such as interpolation, forward filling, or removal depending on the amount and pattern of missing data. Abnormal readings and outliers are also checked so that extremely incorrect sensor values do not affect the model training process. After cleaning, feature normalization or standardization is applied to scale the numerical values into a suitable range. This helps the forecasting model learn efficiently and prevents features with large numerical values from dominating other important features.

3.2.3 Dynamic Graph Construction

In the proposed system, each air quality monitoring station is represented as a node in a graph. The relationship between two stations is represented as an edge. Unlike a static graph, where station relationships remain fixed, this system uses a dynamic graph whose edge weights can change over time based on environmental conditions.

The edge weight between two stations is constructed using multiple factors. The first factor is spatial distance, where nearby stations are generally expected to have stronger influence on each other. The second factor is pollutant correlation, where stations showing similar pollution patterns are connected more strongly. The third and most important factor is wind information. If the wind direction indicates that pollutants can move from one station toward another, the connection between those stations becomes stronger. Wind speed is also considered because stronger wind may increase the movement of pollutants over a larger area.

By combining distance, pollutant similarity, wind direction, and wind speed, the system creates a wind-aware dynamic graph. This graph helps the model understand not only which stations are geographically close, but also which stations are likely to influence each other due to changing wind conditions.

3.2.4 Spatio-Temporal Forecasting Model

The processed data and the dynamic graph are given as input to the forecasting model. The forecasting model is designed to learn both spatial and temporal dependencies. Spatial dependency refers to the relationship between different monitoring stations, while temporal dependency refers to the change in pollutant concentration over time.

Graph-based neural network layers such as Graph Convolutional Network (GCN) or Graph Attention Network (GAT) are used to learn spatial relationships between stations. These layers use the dynamic graph to understand how pollution at one station may influence nearby or downwind stations. After spatial feature extraction, recurrent or sequence-based layers such as GRU or LSTM are used to learn temporal patterns from historical data. These temporal layers help the model understand how pollution levels change over hours or days.

The model is trained using historical air quality and weather data. During training, the predicted pollutant concentration is compared with the actual observed concentration. The prediction error is calculated using loss functions, and the model parameters are updated to reduce this error. After training, the model becomes capable of forecasting future pollutant concentrations for selected stations and time horizons.

3.2.5 Forecast Generation

Once the model is trained, it is used to generate future air quality forecasts. The system takes recent air quality readings, weather features, and current wind conditions as input. Based on these inputs, the dynamic graph is updated and the forecasting model predicts future pollutant concentrations.

The output may include predicted values of pollutants such as $PM_{2.5}$ or PM_{10} for the next few hours or days. These predicted values can also be converted into air quality risk levels such as good, moderate, unhealthy, or hazardous depending on the selected air quality index standard. This makes the output easier to understand for non-technical users.

3.2.6 Explainability Module

In addition to forecasting, the system includes an explainability module to make the model output more interpretable. Air quality forecasting models can be complex, especially when graph neural networks and temporal models are used. Therefore, explainability is important to understand why the model produces a certain forecast.

The explainability module identifies important weather features, influential neighboring stations, and significant time periods that contribute to the prediction. For example, the system may show that high $PM_{2.5}$ concentration at a station is influenced by strong wind flow from a nearby polluted station. Similarly, it may identify wind direction, humidity, or previous pollutant concentration as important factors for a high-pollution event. This helps users understand the reasoning behind the forecast rather than only viewing the final predicted value.

3.2.7 What-If Scenario Analysis

The system also includes a what-if scenario analysis module for decision support. This module allows users to observe how air quality may change under different hypothetical conditions. For example, users may analyze the possible effect of reduced emissions, changes in wind direction, or mitigation measures in a specific area.

By modifying selected input conditions and observing the forecasted output, the system can support environmental planning and policy-related analysis. Although the scenario results are not direct real-world guarantees, they provide useful insights into possible pollution trends and the relative impact of different environmental conditions.

3.2.8 Dashboard and Visualization

The final output of the system is presented through an interactive dashboard. The dashboard displays forecasted pollutant concentration, air quality risk levels, spatial pollution maps, trend graphs, and explanation-based insights. It allows users to view pollution patterns across different monitoring stations and understand how wind may affect pollutant movement.

The dashboard may also include reports and visual summaries for easier interpretation. Researchers can use the dashboard to analyze model behavior, while policymakers and concerned users can use it to understand air quality risk and possible mitigation strategies. Thus, the dashboard acts as the final decision-support interface of the proposed system.

3.2.9 Overall System Flow

Overall, the system starts by collecting air quality and meteorological data, followed by preprocessing and synchronization of the collected records. A dynamic graph is then constructed using station distance, pollutant correlation, and wind information. The spatio-temporal forecasting model uses this graph and historical data to predict future air quality. The explainability and scenario analysis modules provide meaningful interpretation of the results, and the final outputs are displayed through a dashboard for practical decision support.

3.3 Flowchart

Figure 3-2 presents the overall workflow of the proposed methodology.

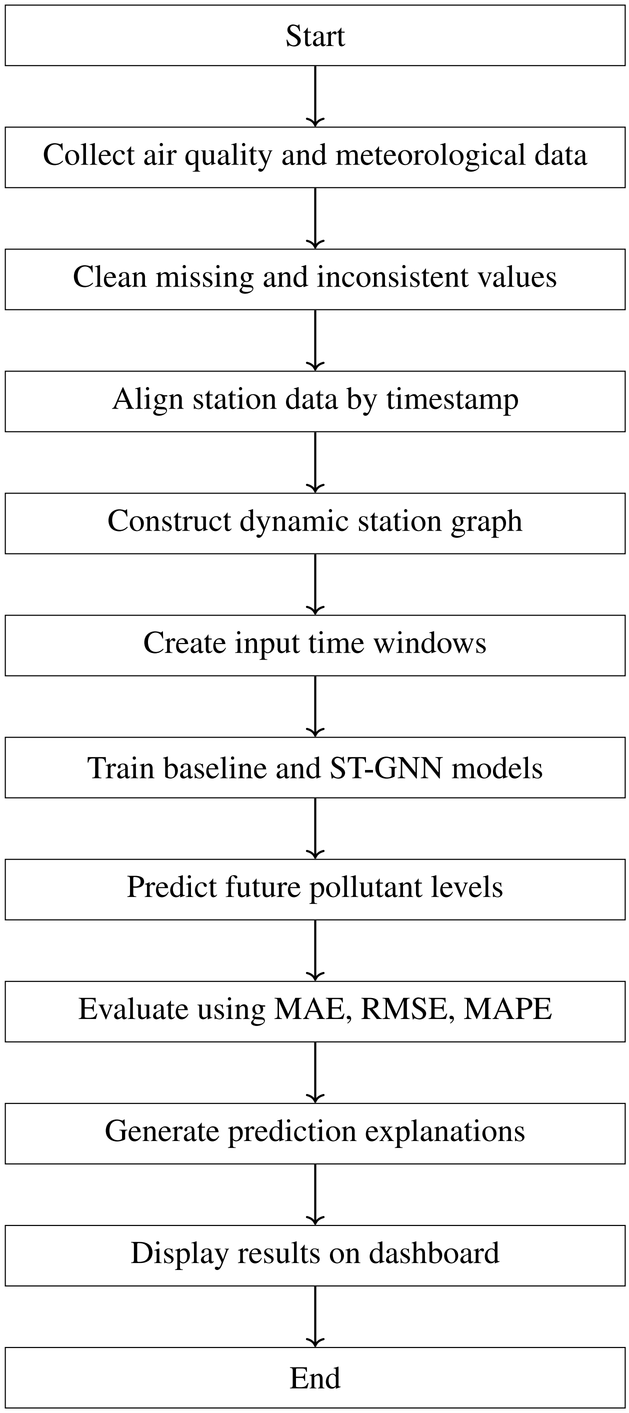

Figure 3-2 Proposed air quality forecasting workflow

3.4 Instrumentation Tools

Table 3-1 Proposed Technical Stack

Category	**Tools and Technologies**
Data Processing	Python, pandas, NumPy, scikit-learn
Deep Learning	PyTorch, PyTorch Geometric, optional Deep Graph Library (DGL)
Forecasting Models	Persistence baseline, regression, LSTM, GRU, GCN+GRU, GAT+GRU
Data Sources	OpenAQ, Open-Meteo, NASA POWER, public benchmark air quality datasets
Backend	FastAPI, Representational State Transfer (REST) services, experiment result APIs
Frontend	React or Next.js, Leaflet or Mapbox, Recharts
Database and Storage	PostgreSQL, local storage for datasets, models, and experiment logs
Deployment	Docker, local server, optional cloud deployment

3.5 Evaluation Methodology

The project will evaluate forecasting models using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), MAPE, and horizon-wise error analysis. The proposed model will be compared with baseline temporal models to check whether dynamic graph learning improves forecasting accuracy. Explainability will be assessed qualitatively by inspecting whether important stations, weather factors, and time windows are consistent with wind direction, distance, and pollutant history. The dashboard will be evaluated based on whether it can clearly present forecasts, risk levels, explanations, and what-if scenario comparisons.

4 EXPECTED OUTCOME

The primary goal of this project is to develop and validate a research-grade prototype for explainable, wind-aware air quality forecasting. The proposed system serves as a decision-support tool by bridging the gap between raw environmental data and actionable policy insights.

4.1 Technical Outcomes

The expected technical outcomes of the project are as follows:

- **Robust Data Pipeline:** A fully automated preprocessing framework for multi-source data ingestion, timestamp synchronization, outlier detection, missing value handling, and normalization to support high-quality model training.
- **Wind-Aware Dynamic Graph Engine:** A dynamic graph construction method that updates station-to-station relationships by integrating spatial distance, pollutant correlations, and wind direction and speed information.
- **Hybrid Forecasting Framework:** A forecasting framework consisting of traditional baseline models such as LSTM and GRU, along with advanced spatio-temporal graph neural network models such as GCN-GRU and GAT-GRU for performance benchmarking.
- **Explainability and Decision Support Module:** A module that interprets model outputs to provide human-readable insights, such as important meteorological features, influential neighboring stations, and temporal factors contributing to high-pollution events.
- **Interactive Analytics Dashboard:** A centralized visualization platform for displaying air quality forecasts, spatial risk maps, historical trends, and interpretability reports for end-users.

5 PROJECT SCHEDULE

The project is planned for approximately one academic year. Work will begin with literature review and dataset selection, followed by data collection, preprocessing, graph construction, baseline models, ST-GNN development, explainability, dashboard development, testing, documentation, and final presentation.

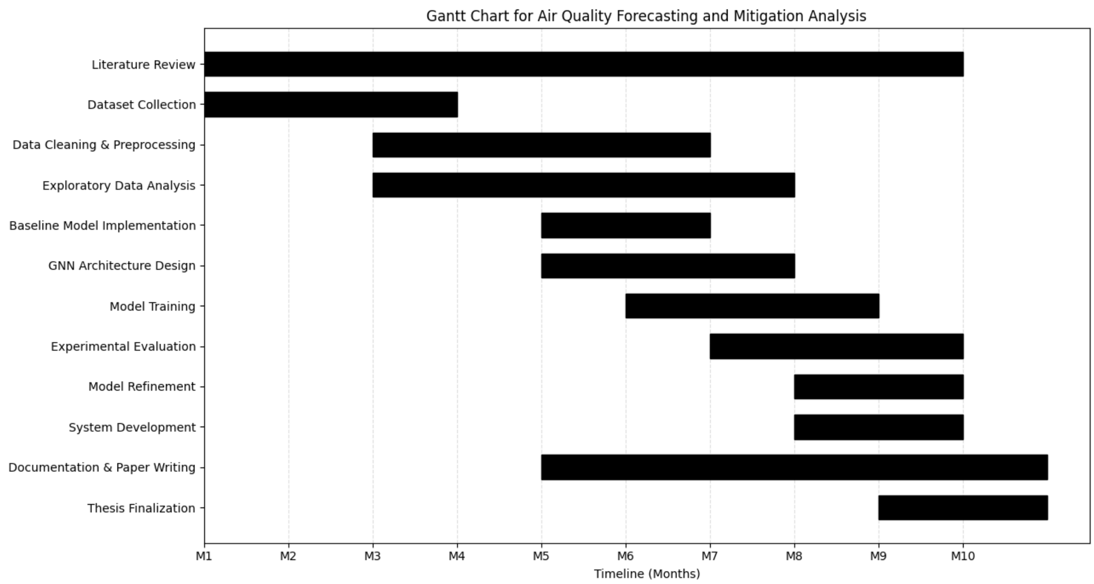

Figure 5-1 Gantt Chart

6 PROJECT BUDGET

The proposed system is mainly a software and research project, so the required budget is relatively low. Most libraries and datasets are open source or publicly available. The main expected expenses are internet access, optional cloud or Graphics Processing Unit (GPU) compute, deployment, documentation, and miscellaneous project needs.

Table 6-1 Estimated Project Budget

Category	**Description**	**Estimated Cost**
Internet and Data Access	Downloading datasets, accessing APIs, and reviewing research resources	NRs. 2,000
Cloud/GPU Usage	Optional Google Colab Pro, Kaggle, or equivalent compute for model training	NRs. 5,000
Domain/Hosting	Demo deployment for dashboard presentation, if needed	NRs. 3,000
Printing and Binding	Proposal, progress report, final report, and presentation materials	NRs. 3,000
Miscellaneous	Backup expenses and unforeseen project needs	NRs. 2,000
	Total Estimated Cost	**NRs. 15,000**

Most software tools used in this project will be open source. GPU usage may be reduced by using free Google Colab or Kaggle Notebook resources during early experiments.

7 FEASIBILITY ANALYSIS

7.1 Technical Feasibility

The project is technically feasible because public air quality datasets, weather APIs, and open-source machine learning frameworks are available. Data can be collected from OpenAQ, Open-Meteo, NASA POWER, or public benchmark datasets. The model can be developed using Python, PyTorch, PyTorch Geometric, pandas, and related tools. A smaller dataset or selected city can be used during the first phase to reduce computational complexity.

7.2 Operational Feasibility

The final system can be operated as a web-based dashboard. Users can view station-wise forecasts, risk levels, explanations, and what-if scenarios. The system is suitable as a research prototype and decision-support tool for academic demonstration. It is not intended to replace official government air quality warning systems.

7.3 Economic Feasibility

The project has low financial cost because most tools and datasets are open-source or publicly accessible. The main possible cost is cloud or GPU usage, which can be minimized using free resources such as Google Colab or Kaggle Notebook. The estimated budget is shown in Table 6-1.

7.4 Schedule Feasibility

The project is feasible within one academic year if developed in phases. The first working prototype can be built using baseline models and one selected dataset. The advanced ST-GNN, explainability, and scenario modules can be added progressively after the data pipeline and baseline models are stable.

7.5 Risk and Mitigation

Table 7-1 Project Risks and Mitigation Strategies

Risk	**Mitigation Strategy**
Missing or inconsistent sensor data	Use interpolation, timestamp alignment, quality checks, and selected stations with reliable coverage.
Limited station density	Start with a city or benchmark dataset that has enough monitoring stations for graph modeling.
Weather data mismatch	Use nearest weather grid points and align meteorological features by timestamp.
Long training time	Start with lightweight baseline models and use optional GPU resources only for selected experiments.
Overfitting	Use validation sets, early stopping, regularization, and comparison with simple baselines.
Weak explanation quality	Compare attention-based explanations with feature importance and physical wind-direction evidence.
Dashboard complexity	Build a simple MVP first, then add maps, scenario comparison, and reporting features.

7.6 Limitations

The system will not replace official air quality warning systems. Prediction accuracy will depend on data quality, station density, and availability of reliable weather data. The mitigation module will be treated as a what-if decision-support tool, not a guaranteed causal policy recommendation.

7.7 Final Feasibility Statement

The proposed project is feasible as a major project because the required data sources, algorithms, and development tools are available. It has sufficient research depth through dynamic graph construction, spatio-temporal forecasting, explainability, and mitigation scenario analysis while remaining practical enough to complete within the academic project period.

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